

DIYARI M. SALIH

M2 researcher at IBISC, Paris-Saclay, working on visual-inertial SLAM and dense mapping with foundation models. First author of **VIDAR** (arXiv:2607.17171). Published at **IEEE ICARM 2026**; reviewer for **IEEE ICARCV 2026**.

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[arXiv:2607.17171](https://arxiv.org/abs/2607.17171)

EXPERIENCE

- IBISC Laboratory, Université Évry Paris-Saclay** Évry, France
Research and Engineering Intern, Visual-Inertial Perception & Mapping Mar 2026 – Sep 2026
 - VIDAR (first-author framework)**: Fused IMU and camera data (SVO + Ceres visual-inertial odometry) with a geometric foundation model (Depth Anything 3) for metric dense monocular mapping; implemented camera calibration, Sim(3)/Umeyama alignment, and pose-graph loop closure in Python and C++.
 - Sensor fusion result**: Adding inertial constraints to monocular visual odometry cut mean trajectory RMSE from **0.898 m to 0.125 m** across 11 EuRoC MAV sequences.
 - Dense reconstruction result**: Reached **0.676** mean dense F@0.10 (above the ground-truth-pose baseline of 0.504) *without* ground-truth poses, reducing metric scale error to **~1%**.
 - Embedded deployment profiling**: Benchmarked latency and memory across GPU classes: lightweight VIO under **1 GB** GPU memory versus 9.5–11 GB for the learned depth model, quantifying the trade-offs.
- Ala Company** Erbil, Iraq
Junior PLC Technician 2024
 - Industrial automation**: PLC diagnostics and wiring, machinery installation, motor and solar-system commissioning, and fault-finding on live equipment.
- Unigaz** Erbil, Iraq
Engineer and Academic Intern, LPG & Hydraulic Infrastructure 2021 – 2023
 - Safety-critical field engineering**: Installation supervision on a 9-building pressurised LPG network: flow-rate regulation, pressure testing, leak detection, AutoCAD routing, and quality-control inspection.

PUBLICATIONS AND PRESENTATIONS

- D. M. Salih**, L. Hu, N. Ait Oufroukh-Mammar, F. Bonardi. “VIDAR: Visual-Inertial Dense Alignment and Reconstruction via a Geometric Foundation Model.” Preprint [arXiv:2607.17171 \[cs.RO\]](https://arxiv.org/abs/2607.17171), July 2026. *Under review at IEEE ICARCV 2026* (invited session). (**First author.**)
- L. Hu, A. Zidan, A. Qawouq, **D. M. Salih**, N. Ait Oufroukh, F. Bonardi. “Can Geometric Foundation Models and Vision SLAM Serve as a Viable Alternative to LiDAR for Indoor Robot Localization?” **IEEE ICARM 2026**, Université Évry Paris-Saclay, France, 21–23 July 2026. *Accepted; camera-ready submitted, to appear in IEEE Xplore*. Unified Gazebo/ROS benchmark of **10 SLAM systems** (LiDAR, feature-based, learning-based, and geometric-foundation-model) across 6 indoor environments on a Quanser QCar (2D LiDAR, RGB-D, wheel odometry); 15,547 frames and 1.04 km of teleoperated trajectory. Contributed the SVO monocular visual-odometry baseline (RGB-only, Sim(3)-aligned ATE) and the vision-versus-LiDAR comparison.
- D. M. Salih**, I. Chaabeni, N. Ait Oufroukh-Mammar. “Identity-Gated Gesture Control and Deterministic Mode Arbitration for Indoor Human-Drone Interaction.” *Submitted July 2026, ARO-The Scientific Journal of Koya University* (Scopus / WoS ESCI, DOAJ Seal); under editorial consideration. (**First author.**) Also presented as a poster at the Workshop on Safety-Critical Software, Control and Learning for Robotics, **IEEE ICARM 2026**. Identity-gated control on a DJI Tello EDU: MediaPipe 21-point hand landmarks with an RBF-SVM classifier (**99.71%** accuracy, macro F1 0.9971 over 7 gestures) and ArcFace-style ONNX face embeddings restricting control to the enrolled operator (**0.32% EER** over 13,410 images). Deterministic mode arbitration under a safety > authorization > control priority order; a local Qwen2.5-0.5B model logs a rationale for every decision *outside* the control loop. Hybrid perception **under 30 ms/frame** ($\sim 9\times$ faster than InsightFace-only). Code: github.com/Diyari-Fariq-M-salih/gesture-controlled-tello-drone.
- L. Hu, **D. M. Salih**, F. Bonardi, N. Ait Oufroukh, R. Ghandour. “GCS-SLAM: Globally Consistent Static Mapping from Dynamic Scenes Without Auxiliary Inputs.” *Manuscript under review* (reviews received; author-response phase). RGB-only monocular dynamic SLAM: segmentation-free dynamic-object filtering, calibration-free tracking, and real-time loop closure, with no intrinsics, depth, or optical-flow inputs.

PROFESSIONAL SERVICE AND AWARDS

- Reviewer, IEEE ICARCV 2026**: 19th Int. Conf. on Control, Automation, Robotics and Vision (IEEE RAS / IEEE CSS technically co-sponsored). Reviewing three submissions in visual-inertial odometry, SLAM, and foundation models for robotic perception.
- Best Volunteer Award, IEEE ICARM 2026**: Recognized by the organizing committee of the [11th IEEE International Conference on Advanced Robotics and Mechatronics](#), hosted by IBISC at Université Évry Paris-Saclay, 21–23 July 2026, for venue setup, graphics, and event operations across the conference week.

EDUCATION

- **Université Paris-Saclay / Université Évry Val-d'Essonne** Évry, France
M2 EEEA, Smart Aerospace and Autonomous Systems 2025 – 2026 (diploma 2027)
 - **Notable modules:** Individual Project **17.5/20**, Flight Planning **16.85/20**, Aerial Robots 14.5/20, AI and Aerospace Systems 14.35/20, Advanced Artificial Perception 14.1/20, Sensor Fusion 14/20.
- **Poznań University of Technology** Poznań, Poland
Double degree MSc, Automatic Control and Robotics 2025 – 2026
 - **Self-initiated robotics builds (team):** *Holonomic 4WD UGV:* chassis fabrication, HW-130 motor-driver integration, control firmware in Arduino C++, stable wireless teleoperation beyond **30 m**. *Two-wheel self-balancing robot:* inverted-pendulum stabilisation from MPU-6050 accelerometer/gyroscope fusion feeding a **PID** loop onto DC motors.
 - **Coursework:** Smart Systems, Fundamentals of Autonomous Systems, Sensor Integration, Adaptive Control.
- **Salahaddin University-Erbil** Erbil, Iraq
B.Sc. Mechanical and Mechatronics Engineering 2019 – 2023
 - **Result:** Graduated **2nd in department** of 39, July 2023; capstone (AcciMap accident analysis of LPG fire scenarios) rated *Excellent*.

TECHNICAL SKILLS

- **Programming:** C++, Python (primary); MATLAB, Bash.
- **Perception & Estimation:** Visual-inertial odometry (VIO), IMU fusion, Vision SLAM (SVO), monocular depth estimation, dense 3D reconstruction, dynamic-scene SLAM and static mapping; feature matching and stereo correspondence, Sim(3)/Umeyama alignment, pose-graph loop closure, ATE evaluation, particle filters, path planning (A*, RRT).
- **Systems & Tools:** Linux (development and embedded workflows), ROS 1/2, Gazebo, CUDA, OpenCV, Ceres, Git, CMake, Jupyter, LaTeX.
- **Machine Learning:** PyTorch, scikit-learn; CNNs, LSTMs, SVMs; object and keypoint detection (MediaPipe); geometric foundation models; ONNX Runtime deployment; benchmarking and latency/memory profiling.
- **Sensors & Integration:** IMUs (MPU-6050), RGB and RGB-D / depth cameras, 2D LiDAR, wheel odometry; acquisition and data logging (rosbag), time synchronisation, intrinsic and extrinsic calibration, UART/serial, USB, Arduino and microcontroller interfacing.
- **Control:** PID and cascaded loops, proportional/feed-forward control, trajectory generation, state machines and mode arbitration, adaptive control.
- **CAD & Fabrication:** SolidWorks, AutoCAD; 3D printing (Ender-3, custom Marlin firmware), resin printing, STL modelling, 3D scanning.

CERTIFICATIONS, LANGUAGES AND ACTIVITIES

- **Languages:** English (IELTS 8.0, C1; EF SET C2), French (A2), German (A2), Arabic (native), Kurdish (native).
- **Certifications:** German Summer School on Additive Manufacturing (TU Bergakademie Freiberg, DAAD DiLAM Project, Sep 2021); Front-End Design and Development Bootcamp, 117 hours (Rwanga Foundation / WeCode, 2022); Introduction to UX/UI (Recoded, 2023); HSE Certified Expert and HSE training (Canadian Business Academy, 2022).
- **Activities:** Member, Évry Robotics Association (EVOROBS), 2025–2026: hexapod trajectory and control with an on-device Gemma 3 1B model; technical collaborator on an RGB-D assistive navigation system for visually impaired users (Poznań University of Technology, 2026): algorithm design, and safety analysis of depth-based obstacle detection and command arbitration; Team Manager, IREX Global Solutions Challenge, 2022–2023.

REFEREES

Dr. Naima Ait Oufroukh-Mammar (Head of Department, IBISC Laboratory), naima.aitoufroukh@univ-evry.fr
Dr. Marcin Kielczewski (Poznań University of Technology), marcin.kielczewski@put.poznan.pl
Lingxiang Hu (Research Co-supervisor, IBISC Laboratory), hulxhlx@gmail.com